Deep learning is a specialized subset of machine learning that uses multi-layered artificial neural networks to mimic the human brain and automatically learn complex patterns from raw, unstructured data. [1, 2, 3, 4]

How Deep Learning Works

- **Artificial Neural Networks (ANNs):** Models are built from layers of interconnected nodes or neurons. Data moves from an input layer, through multiple hidden layers, to an output prediction. [1, 2, 3]
- **Automatic Feature Extraction:** Unlike traditional machine learning, deep learning models discover features (like edges or shapes in images) on their own without human intervention. [1, 2]
- **Training via Backpropagation:** Networks adjust connection weights and biases using error gradients and gradient descent to optimize performance. [1, 2]

Common Architectures

- **Convolutional Neural Networks (CNNs):** Designed for spatial data like computer vision, image classification, and object recognition. [1, 2]
- **Recurrent Neural Networks (RNNs) & LSTMs:** Built to process sequential data, time-series, and speech recognition. [1, 2]
- **Transformers:** Rely on self-attention mechanisms to power modern generative AI and large language models. [1, 2]

Key Applications

- **Computer Vision:** Self-driving cars, facial recognition, and medical image analysis. [1, 2]
- **Natural Language Processing (NLP):** Machine translation, automated text generation, and sentiment analysis. [1]
- **Speech Recognition:** Voice assistants and real-time audio transcription. [1, 2]

Watch this introductory video to learn more about the structure and working of deep learning networks: